

CustomsGuard

Autonomous AI engine for commercial invoice auditing,
tariff classification, and container detention risk prevention.

IBM SkillsBuild x Hacktiv8 National Hackathon 2026

Presenter: Rafif Dzakwan Muchsin

Daftar Isi

Langkah strategis transformasi verifikasi kepabeanan manual menuju audit otonom berbasis AI untuk mengeliminasi risiko penahanan kargo di pelabuhan.

01 Gambaran Umum

02 Permasalahan

03 Solusi

04 Agen AI

05 Dampak

07 Analisis Pasar

Gambaran Umum

Jalur laut mendominasi lebih dari 60% perdagangan Asia Tenggara, namun lebih dari 76% proses kepabeanan nasional masih diperiksa secara manual berbasis kertas.

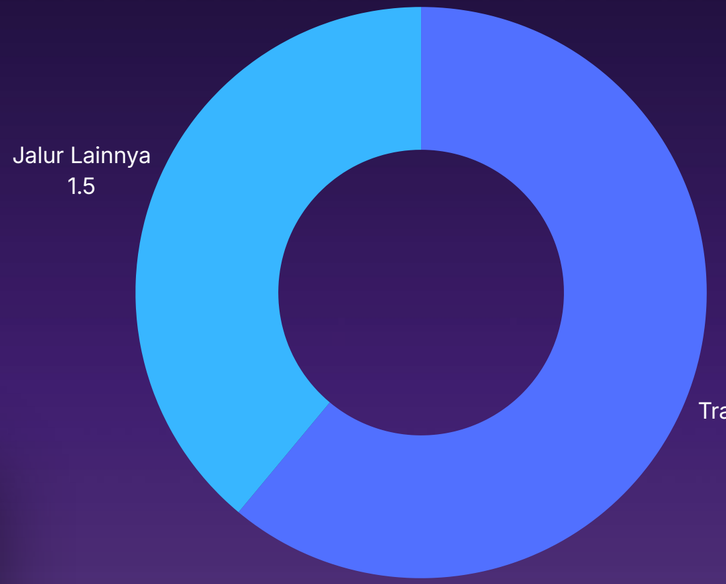
Proses lambat ini memicu penumpukan kontainer dan risiko denda pelabuhan. CustomsGuard hadir mengotomatisasi audit kepatuhan agar verifikasi dokumen berjalan instan dan presisi.

- Agensi PPJK independen yang terbebani verifikasi manual kode HS hingga 30 menit per item.
- Freight forwarder dan 3PL yang terancam lonjakan denda demurrage akibat pemeriksaan Jalur Merah.
- Petugas kepatuhan importir korporasi yang kerap luput mengklaim restitusi kelebihan bayar bea masuk.

>76%

Operasional kepabeanan di Indonesia masih mengandalkan pemeriksaan dokumen manual berbasis kertas (Mordor Intelligence, 2025).

Proporsi Jalur Perdagangan Barang ASEAN (Triliun USD)



Merujuk data ASEAN Statistical Yearbook 2023, perdagangan barang ASEAN menembus 3,85 triliun dolar AS (2,35 triliun dolar AS melalui jalur laut), sementara sektor transportasi dan logistik Indonesia menyumbang 983,5 triliun rupiah terhadap PDB nasional.

Permasalahan

Kongesti Pelabuhan

Data operasional PT Pelindo Regional 2 mencatat Pelabuhan Tanjung Priok melayani 8,3 juta TEUs sepanjang tahun 2025. Muatan tanpa slot janji temu atau bermasalah dokumen mengantre 7 hingga 10 hari sebelum kliring selesai menurut Mordor Intelligence.

Denda Demurrage

Sesuai tarif resmi CMA CGM Indonesia, importir hanya memperoleh 5 hari masa bebas (free days). Pada hari ke-6 hingga ke-10, biaya denda mencapai 101 dolar AS per hari untuk kontainer kering standar ukuran 40 kaki.

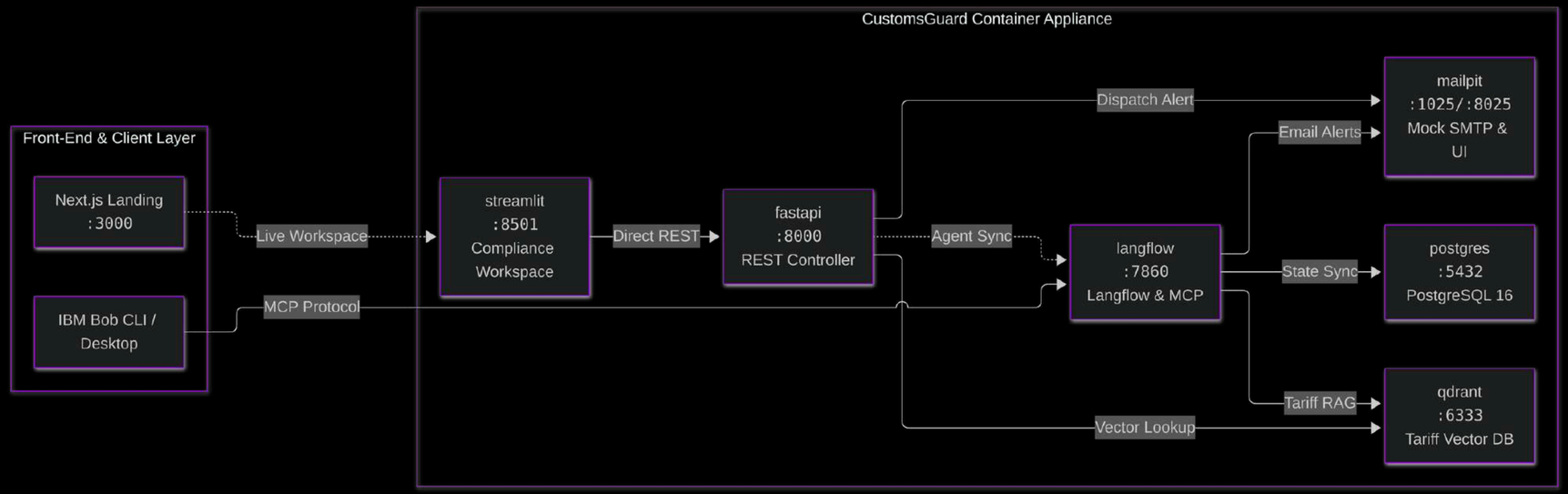
Regulasi Lartas

Berdasarkan basis data ITC MAcMap WTO dan UNCTAD, 546 subpos HS 6 digit wajib memiliki izin edar atau persetujuan impor awal (seperti SDPPI Kominfo, Kemenkes, atau BPOM). Dokumen yang tidak lengkap menahan kargo tanpa batas waktu pasti.

Sanksi Administratif

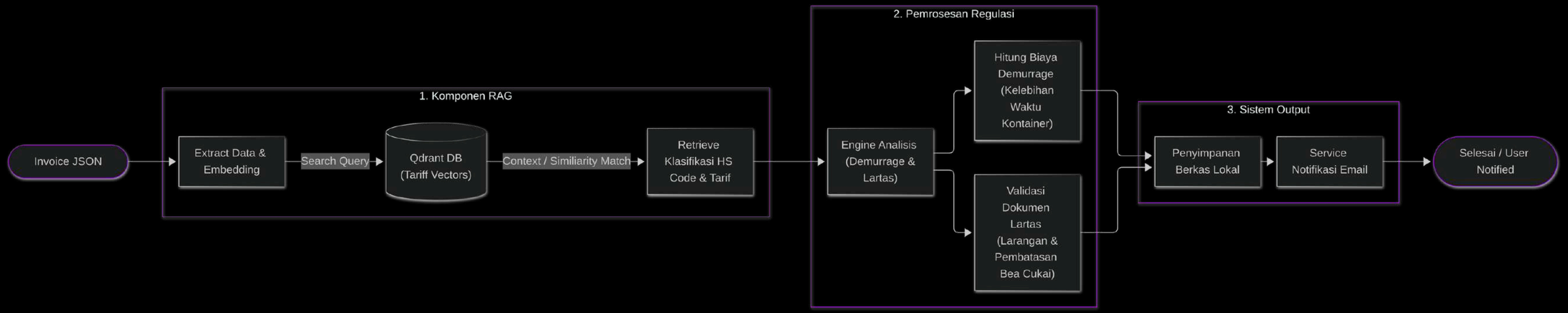
Pasal kepabeanan dalam Undang-Undang Nomor 17 Tahun 2006 menetapkan bahwa kekurangan pembayaran bea masuk akibat kesalahan penetapan tarif dikenai denda administratif sebesar 100 persen hingga 1.000 persen.

High-Level Architecture



1. Dual Architecture: audit kilat via REST API dan asisten cerdas interaktif via IBM Bob.
2. Integrasi IBM Bob dan Langflow menggunakan standar terbuka Model Context Protocol (MCP).
3. Penyimpanan vektor Qdrant lokal untuk pencarian tarif dan PostgreSQL untuk status agen.
4. Seluruh sistem berjalan mandiri di kontainer Docker lokal (air-gapped) demi privasi data penuh.

Main Agentic Workflow



1. Pencarian vektor semantik terhadap 5.612 subpos tarif HS 6 digit Indonesia pada Qdrant lokal.
2. Skema perhitungan tarif demurrage progresif sesuai aturan resmi CMA CGM Indonesia.
3. Menghitung selisih tarif untuk mengidentifikasi kelebihan deklarasi bea sebagai peluang pengembalian pajak.
4. Pengarsipan berkas audit ke penyimpanan lokal serta pengiriman peringatan email real-time.

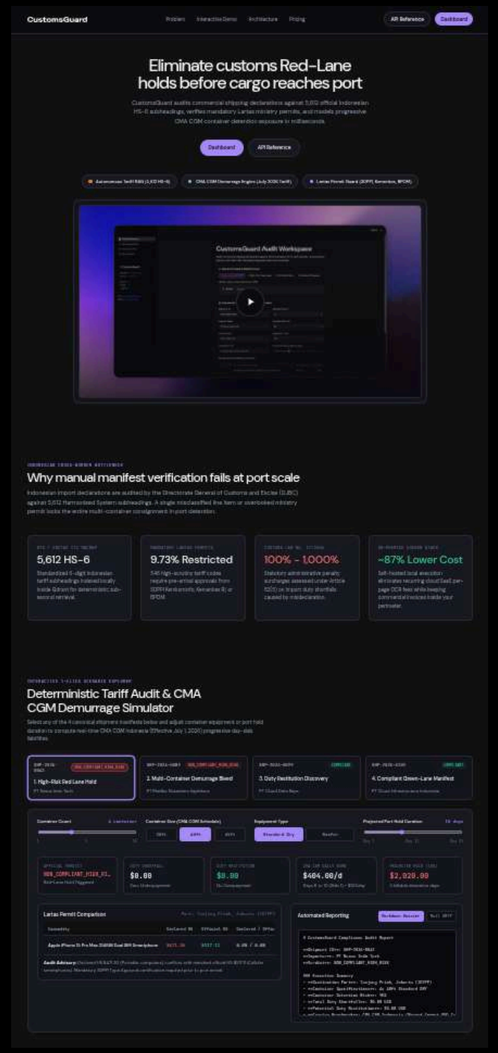
Agen AI

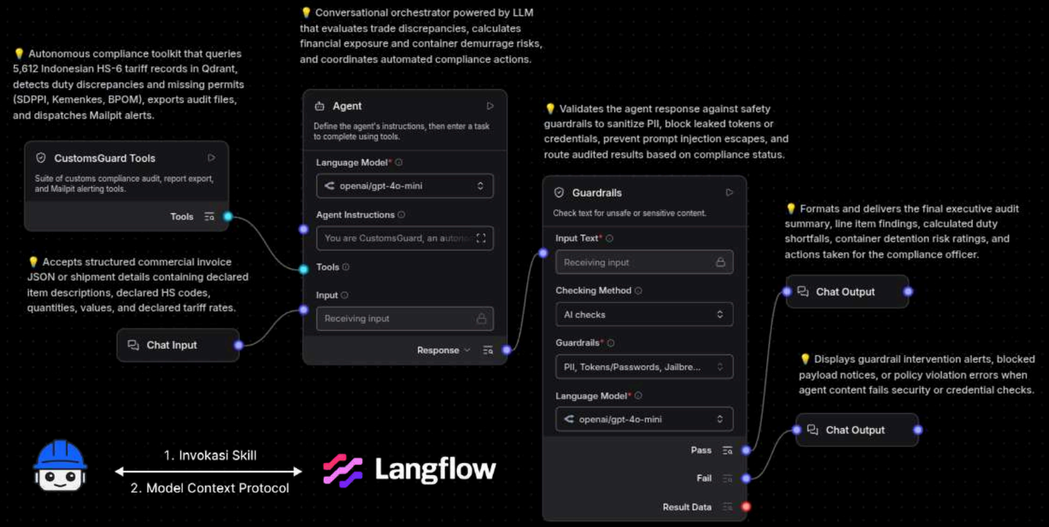
- Langflow Tools
  - RAG
  - Audit Regulasi
  - Report & Dispatch

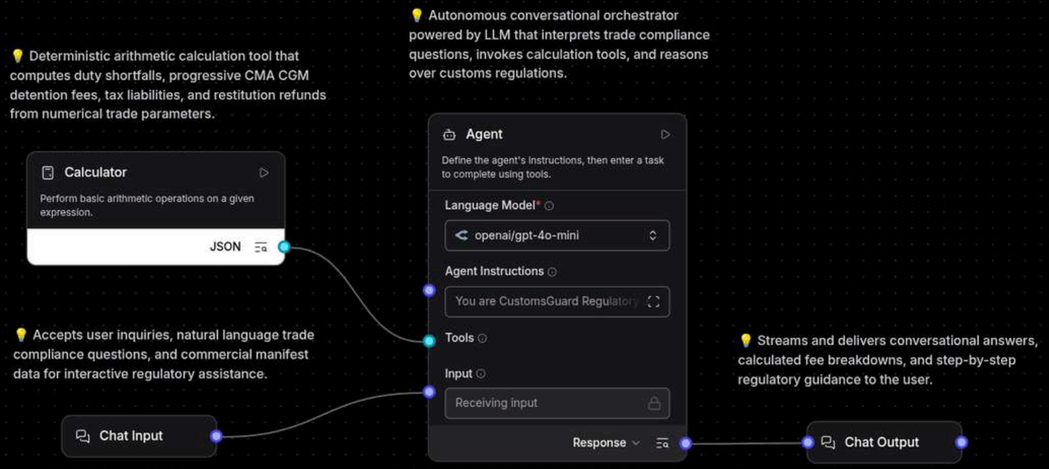
- Protokol MCP & Bob Skill
- Guardrail Informasi
- NextJS Landing

- Streamlit Dashboard
  - Audit Workspace
  - Demurrage Matrix
  - Dossier Archive
  - Bob Assistant

- E2E local-first
  - Deployment Docker
  - LLM agnostic







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HACKTIV8

CustomsGuard Audit Workspace

Audit commercial shipping declarations against 5,612 Indonesian HS-6 tariff records, Lartas ministry permits, and CMA CGM Indonesia progressive detention schedules.

Scenario Presets & Manifest Input

1. Red-Lane Hold (SDPPI) | 2. Multi-Cntr Demurrage | 3. Duty Restitution | 4. Compliant Clearance

Upload custom commercial invoice JSON

Upload — 200MB per file • JSON

Commercial Invoice & Container Parameters

Shipment ID: SHP-2026-0042	Container Count: 4
Importer Name: PT Nexus Indo Tech	Container Size (ft): 40
Ocean Carrier: CMA CGM / APL	Equipment Type: DRY
Destination Port: Tanjung Priok, Jakarta (IDTPP)	Projected Hold Duration (Days): 10

Declared Commodity Line Items

ID	Commodity Description	Declared HS	Qty
1	Apple iPhone 15 Pro Max 256GB Dual SIM Smartphone	8471.30	100

Run Autonomous Compliance Audit

Executive Financial & Liability Summary

Official Verdict	Total Duty Shortfall	Duty Restitution Recovery
HIGH RISK — Red-Lane Hold	$0.00 — Zero Shortfall	$0.00 — No Overpayment

CMA CGM Daily Burn Rate	Projected Hold Liability (10 Days)
$404.00 / day — Days 6 to 10 ($101.00/day)	$2,020.00 — 5 Billable Detention Days (7d Free)

Executive Synthesis & Discrepancy Alerts

Shipment SHP-2026-0042 for importer **PT Nexus Indo Tech** (4x 40ft Standard Dry via CMA CGM / APL into Tanjung Priok, Jakarta (IDTPP)) received official compliance verdict **NON_COMPLIANT_HIGH_RISK**.

Red-Lane customs hold and container detention risk is **ACTIVE**. Total duty shortfall is **$0.00 USD**, with an active CMA CGM daily burn rate of **$404.00 USD/day** (Days 6 to 10 ($101.00/day)) and a projected 5-day detention liability of **$2,020.00 USD**.

Critical Compliance & Regulatory Notes:

- Declared HS 8471.30 conflicts with matched HS 8517.11
- Missing mandatory regulatory permits: Telecommunication Type Approval / SDPPI Certification (Kemkominfo)
- CMA CGM Demurrage & Detention risk active: 40ft Standard Dry (Days 6 to 10 ($101.00/day)). Consignment burn rate: $404.00/day ($101.00/day/container). Projected 5-day hold liability: $2,020.00.

Dossier Path: /app/reports/2026-10-24/SHP-2026-0042/audit_report.md | SMTP Alert: Alert won't queue to Mailpit for localhost:8025/api/v1/messages/send...

Itemized HS-6 & Lartas Permit Comparison

ID	Commodity Description	Declared HS	Official HS	Declared Duty	Official Du
1	Apple iPhone 15 Pro Max 256GB Dual SIM Smartphone	8471.30	8517.11	0.0%	0.0%

CMA CGM Progressive Demurrage Matrix

Official CMA CGM Indonesia Merged Import Demurrage & Detention Tariff Schedule (Effective July 1, 2026) evaluated for the active shipment.

Active Equipment Profile

- **Shipment ID:** SHP-2026-0042
- **Carrier Benchmark:** CMA CGM Indonesia (Merged Import D&D Tariff)
- **Equipment:** 4x 40ft Standard Dry
- **Free Time Allowed:** 5 Days

Detention Exposure

- **Detention Risk:** ACTIVE
- **Active Tier:** Days 6 to 10 ($101.00/day)
- **Fleet Burn Rate:** $404.00 USD/day
- **Cumulative Hold (10d):** $2,020.00 USD

Day-Slab Liability Breakdown

Tariff Tier	Equipment	Rate / Cntr / Day	Fleet Burn / Day	Active Days	Tier Subtotal
Days 1 to 5 (Free)	4x 40ft Dry	$0.00	$0.00	5	$0.00
Slab 1: Days 6 to 10	4x 40ft Dry	$101.00	$404.00	5	$2020.00
Slab 2: Days 11 to 15	4x 40ft Dry	$121.00	$484.00	0	$0.00
Slab 3: Days 16 to 20	4x 40ft Dry	$151.00	$604.00	0	$0.00
Slab 4: Days 21+	4x 40ft Dry	$161.00	$644.00	0	$0.00

Evidentiary Dossier Archive

Browse timestamped compliance dossiers stored in ./data/reports and export official Markdown or JSON evidence packages.

Archived Shipment Audits

Date	Shipment ID	Importer	Verdict	Cntr	Shortfall	Restitution
2026-10-24	SHP-2026-0042	PT Nexus Indo Tech	HIGH RISK	4	$0.00	$0.00
2026-10-23	SHP-2026-0089	PT Medika Nusantara Sejahtera	DISCREPANCY	3	$6000.00	$0.00
2026-10-23	SHP-2026-0042	PT Nexus Indo Tech	HIGH RISK	4	$0.00	$0.00
2026-10-22	SHP-2026-0105	PT Cloud Infrastructure Indonesia	COMPLIANT	1	$0.00	$0.00
2026-10-22	SHP-2026-0091	PT Cloud Data Raya	COMPLIANT	1	$0.00	$15000.00
2026-10-22	SHP-2026-0089	PT Medika Nusantara Sejahtera	DISCREPANCY	2	$6000.00	$0.00
2026-10-22	SHP-2026-0042	PT Nexus Indo Tech	HIGH RISK	4	$0.00	$0.00
2026-09-28	SHP-2026-0042	PT Nexus Indo Tech	HIGH RISK	4	$0.00	$0.00
2026-09-27	SHP-2026-0089	PT Medika Nusantara Sejahtera	DISCREPANCY	3	$6250.00	$0.00
2026-09-27	SHP-2026-0042	PT Nexus Indo Tech	HIGH RISK	4	$0.00	$0.00

Select Archived Dossier to Inspect & Export

2026-10-24 | SHP-2026-0042 (HIGH RISK)

Download Markdown (SHP-2026-0042.md) | Download JSON (SHP-2026-0042.json)

CustomsGuard Compliance Audit Report

Shipment ID: SHP-2026-0042
Generated At: 2026-10-24T04:39:48.026218
Verdict: NON_COMPLIANT_HIGH_RISK

Executive Summary

- **Importer:** PT Nexus Indo Tech
- **Destination Port:** Tanjung Priok, Jakarta (IDTPP)
- **Container Count:** 4 (40ft Standard Dry)
- **Container Detention Risk:** YES
- **Total Duty Shortfall:** $0.00 USD
- **Potential Duty Restitution Savings:** $0.00 USD
- **Carrier Benchmark:** CMA CGM Indonesia (Merged Import D&D Tariff)
- **Free Time Allowed:** 5 Days Free
- **Estimated Daily Demurrage:** $404.00 USD / day ($101.00/day per container, Days 6 to 10 ($101.00/day))
- **Projected 5-Day Hold Liability:** $2,020.00 USD

Critical Discrepancies & Regulatory Violations

1. Declared HS 8471.30 conflicts with matched HS 8517.11
2. Missing mandatory regulatory permits: Telecommunication Type Approval / SDPPI Certification (Kemkominfo)
3. CMA CGM Demurrage & Detention risk active: 40ft Standard Dry (Days 6 to 10 ($101.00/day)). Consignment burn rate: $404.00/day ($101.00/day/container). Projected 5-day hold liability: $2,020.00.

Itemized Audit Findings

Item ID	Declared Description	Declared HS	Matched HS	Declared Duty	Officia
1	Apple iPhone 15 Pro Max 256GB Dual SIM Smartphone	8471.30	8517.11	0.0%	0.0%

Automated Actions Taken

- Audit dossier generated on local disk at /app/reports/2026-10-24/SHP-2026-0042/audit_report.md .
- Urgent detention alert email queued in Mailpit (http://localhost:8025).

IBM Bob Regulatory Q&A Assistant

Stateless one-shot trade compliance assistant powered by the bounded IBM Bob CLI worker pool (240s hard timeout with local HS-6 RAG & CMA CGM fallback).

Regulatory Query Console

PER-8517.11 Smartphone Permits | MS-9018.12 Medical Scanners | CMA CGM July 2026 Slabs

Ask a trade compliance, HS-6 classification, Lartas permit, or CMA CGM demurrage question:

Explain the CMA CGM Indonesia July 1, 2026 progressive demurrage slabs for Dry and Reefer containers.

Submit Regulatory Query

✓ Completed via langflow-regulatory-agent (ONLINE in 8402 ms)

Channel: langflow-regulatory-agent | Status: online | Timeout Limit: 240s | Latency: 8402 ms

The CMA CGM Indonesia's import demurrage and detention tariff, effective July 1, 2026, includes specific progressive slabs for both Dry and Reefer containers. The details are summarized below:

Dry Equipment Demurrage Rates

- **Free Days:** 5 days

Days	20ft Container	40ft Container	45ft Container
Days 1-5	$0.00/day	$0.00/day	$0.00/day
Days 6-10	$66/day	$101/day	$141/day
Days 11-15	$86/day	$121/day	$171/day
Days 16-20	$96/day	$151/day	$181/day
Days 21+	$106/day	$161/day	$191/day

Reefer Equipment Demurrage Rates

- **Free Days:** 3 days

Days	20ft Container	40ft Container
Days 1-3	$0.00/day	$0.00/day
Days 4-5	$89/day	$141/day
Days 6-7	$126/day	$176/day
Days 8+	$130/day	$180/day

Key Points

- **Dry Equipment:** After the fifth free day, charges progressively increase based on the duration of the delay in container return.
- **Reefer Equipment:** A shorter grace period of three free days applies, followed by increasing charges that vary based on the days past the free allocation.

General Considerations

- Make sure to plan your logistics flow efficiently to avoid steep demurrage fees, especially after the free days expire.
- It's crucial to be aware of these rates ahead of time to manage your import operations within the allowed timelines effectively.

If you have any questions regarding specific classifications related to customs or need further information, feel free to ask!

Audit the following commercial invoice for shipment SHP-2026-0042. Detect HS code misclas... 21.7k / 270.0k 0.002

Ran Customsguard (ff-bmnb)

CustomsGuard Audit — SHP-2026-0042 · NON-COMPLIANT / HIGH RISK

Shipment Overview

Field	Value
Shipment ID	SHP-2026-0042
Invoice	INV-2026-APL-091
Importer	PT Nexus Indo Tech
Exporter	Shenzhen Global Electronics Ltd
Carrier	CMA CGM / APL
Destination	Tanjung Priok, Jakarta (IDTPP)
Containers	4× 40ft DRY
Projected Hold	10 days
Verdict	NON_COMPLIANT_HIGH_RISK

HS Code Misclassification — Item 1

Field	Declared	Corrected
Description	Apple iPhone 15 Pro Max 256GB Dual SIM Smartphone	—
HS Code	8471.30 (Portable ADP machines)	8517.11 (Telephones for cellular networks / smartphones)
Duty Rate	0.0%	0.0%
Duty Shortfall	$0.00	—

Finding: The iPhone 15 Pro Max is a smartphone and must be classified under **HS 8517.11** ("Telephones for cellular networks or for other wireless networks"). The declared code **8471.30** covers portable automatic data processing machines (laptops/tablets). While the duty rate is coincidentally identical (0%) in this case, the misclassification is a **customs infraction** that can trigger cargo holds, re-examination orders, and penalties under Indonesian customs law (Law No. 17/2006).

SDPPI / Regulatory Permit Status

Permit	Authority	Status
Telecommunication Type Approval / SDPPI Certification	Kemkominfo (Ministry of Communications)	❌ MISSING — REQUIRED

Finding: All smartphones entering Indonesia require a valid **SDPPI (Direktorat Jenderal Sumber Daya dan Perangkat Pos dan Informatika) type approval certificate** issued by the Directorate General of Resources and Equipment of Posts and Informatics (Kemkominfo). Importing 100 units of the iPhone 15 Pro Max without this certification violates **Permenkominfo No. 18/2014** and subsequent regulations. Cargo can be seized at port and the importer faces risk of import ban.

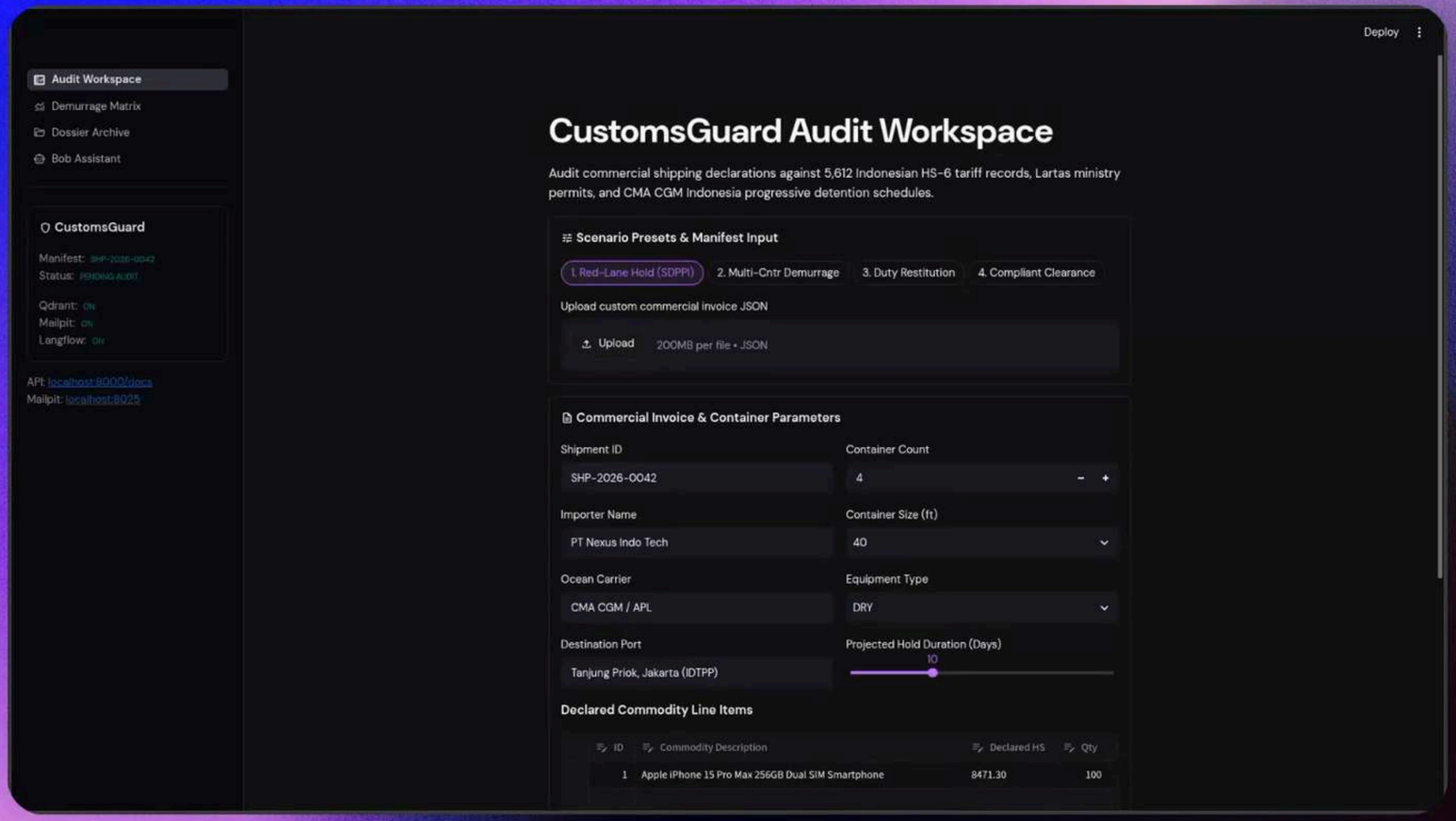
Container Detention Liability (CMA CGM Schedule)

Metric	Value
Free Days	5 days (CMA CGM Indonesia standard)
Billable Days (Days 6–10)	5 days
Rate per Container (Days 6–10)	$101.00/day
Containers	4
Active Daily Burn Rate	$404.00/day
Projected 10-Day Liability	$2,020.00

⚠ The cargo cannot be released until the SDPPI permit is presented and the HS code declaration is corrected. Every additional day of hold adds **$404.00** to the detention bill.

Financial Impact Summary

Category	Amount (USD)
Duty Shortfall	$0.00
Overpayment Refund Eligible	$0.00
Projected Detention Liability	$2,020.00



~1 menit video demo

Jika video tidak berjalan, silahkan akses disini

Peluang & Dampak

Peluang Pasar: Membidik pasar jasa kepabeanan Indonesia senilai USD 2,23 Miliar (dari total freight forwarding ASEAN USD 31,6 Miliar) dengan adopsi digital tumbuh pesat 15,34% CAGR.

Dampak Terukur: Memangkas waktu audit dokumen di Pelabuhan Tanjung Priok dari 4 jam menjadi < 5 detik (>98% efisiensi), mengeliminasi risiko denda demurrage hingga USD 2.020 per insiden penahanan, serta membuka potensi pengembalian pajak lewat fitur restitusi.

Struktur Monetisasi

Starter ($129/bln)

PPJK independen dan forwarder berskala kecil. Kuota 250 audit/bulan, pencarian 5.612 kode HS, verifikasi perizinan Lartas (SDPPI, Kemenkes, BPOM), dan peringatan email otomatis.

Professional ($499/bln)

Menyasar 3PL dan trading house menengah. Kuota 2.500 audit/bulan, multi-seat, kalkulator demurrage progresif CMA CGM, analisis restitusi lebih bayar tarif, dan akses REST API langsung.

Enterprise ($1,499/bln)

Menyasar forwarder multinasional dan operator terminal. Kuota audit tanpa batas, implementasi 100% on-premise Docker (air-gapped) di server lokal, integrasi ERP internal, dan SLA prioritas.

API ($0,75/audit)

Skema pay-as-you-go fleksibel untuk importir berkala atau integrasi software logistik eksternal via REST API dan Model Context Protocol (MCP).

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Pengembalian Investasi

Mencegah satu kali saja penahanan kontainer selama 5 hari pada pengapalan 4 boks ukuran 40 kaki menghemat denda demurrage pelayaran sebesar $2.020 USD berdasarkan skema resmi CMA CGM.

Di samping itu, menghindari kekeliruan penetapan pos tarif senilai $5.000 USD melindungi perusahaan dari denda kepabeanan 100% hingga 1.000% ($5.000 hingga $50.000 USD) sesuai ketentuan Undang-Undang Kepabeanan.

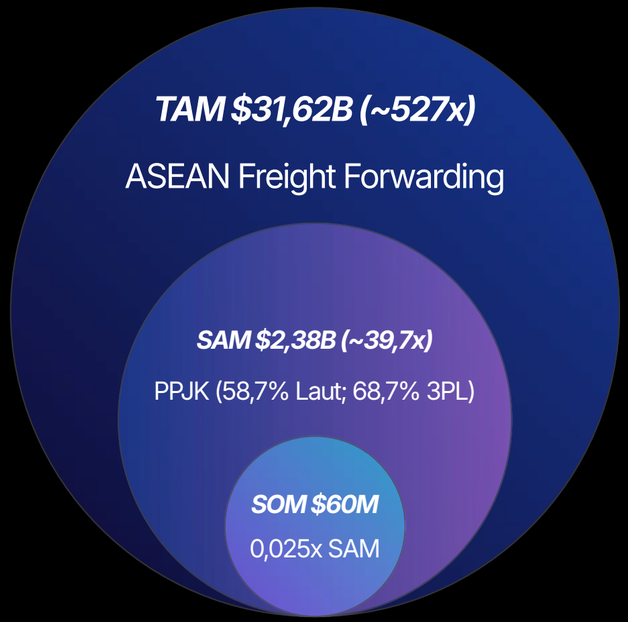
- Memangkas 2-4 jam pemeriksaan manual per manifes menjadi hitungan detik secara konkuren.
- Mengotomatisasi verifikasi kepatuhan lartas lintas kementerian secara instan sebelum kontainer tiba.
- Mendeteksi kelebihan bayar bea masuk secara otomatis untuk peluang restitusi pajak.

~$7,000

Efisiensi biaya infrastruktur tahunan arsitektur mandiri dibanding sewa cloud penuh untuk managed vector database, komputasi VM, dan transfer data.

TAM, SAM, SOM

Metrik Pasar	Nilai & Estimasi	Konteks & Dinamika Operasional	Referensi
Arus Dagang Makro	USD 3,85 Triliun (USD 2,35 T lewat laut); PDB Transportasi RI Rp 983,5 T	Volume kargo laut masif melintasi koridor kepabeanan Asia Tenggara.	ASEAN Statistical Yearbook
TAM (Total Addressable Market)	USD 31,62 Miliar (2025) tumbuh ke USD 41,86 Miliar (2031), CAGR 4,78%	Pasar Freight Forwarding ASEAN (bagian dari total industri logistik ASEAN senilai USD 288 Miliar - 406 Miliar).	Mordor Intelligence / Research and Markets
SAM (Serviceable Addressable Market)	USD 2,23 Miliar (2025) tumbuh ke USD 3,32 Miliar (2031), CAGR 6,83%	Pasar Jasa Kepabeanan (PPJK) Indonesia; 58,67% kargo laut dan 68,74% dikelola oleh forwarder/3PL.	Mordor Intelligence
Realitas Pelabuhan (Tanjung Priok)	8,30 Juta TEU (5,69 Juta boks peti kemas) pada tahun 2025	Beban tinggi Pelabuhan Tanjung Priok; kargo yang bermasalah dokumen tertahan 7 hingga 10 hari.	Pelindo Regional 2 Tanjung Priok
SOM (Serviceable Obtainable Market)	USD 30 Juta hingga USD 60 Juta (Target Pasar Awal)	Membidik 2,5% hingga 5% transaksi pabean berbasis digital dan API di Tanjung Priok (tumbuh 15,34% CAGR).	(Analisis berdasarkan data terkait)



Analisis SWOT

Strength

Audit kepatuhan instan (< 5 detik) berbasis 5.612 kode HS resmi dan 546 izin Lartas. Sistem berjalan 100% lokal (on-premise/air-gapped) sehingga data rahasia invoice terjamin aman, serta dilengkapi fitur unik pendeteksi restitusi lebih bayar bea masuk.

Weakness

Input dokumen saat ini masih berbasis teks atau data JSON terstruktur (belum terintegrasi modul OCR multimodal untuk memindai dokumen fisik secara langsung), serta kalkulator denda penahanan kontainer baru memodelkan satu pelayaran utama (CMA CGM).

Opportunity

Pasar perantara pabean Indonesia bernilai USD 2,23 Miliar didorong arus 8,3 juta TEU di Tanjung Priok dan mandatori digitalisasi CEISA 4.0. Tingginya risiko antrean 7 hingga 10 hari menciptakan kebutuhan mendesak bagi forwarder untuk menggunakan alat audit otomatis.

Threat

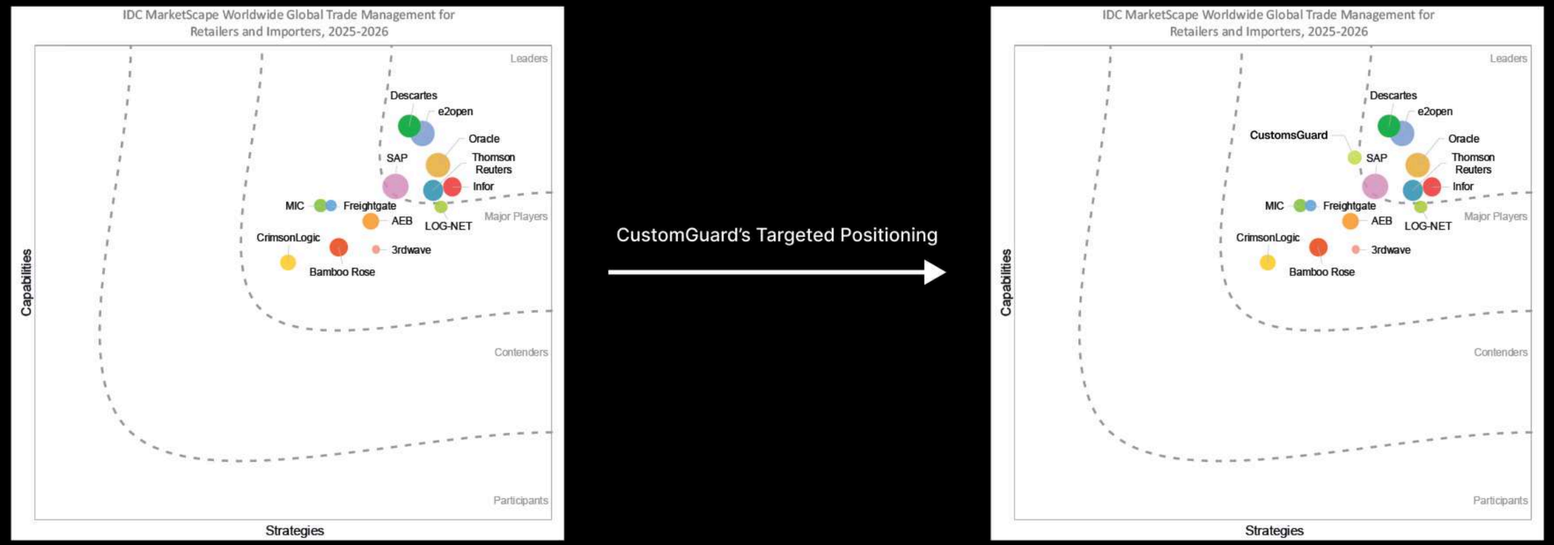
Kebiasaan kerja staf operasional pabean konvensional yang resisten beralih ke otomatisasi AI, perubahan regulasi izin Lartas antar-kementerian yang sangat dinamis, serta potensi vendor EDI petahana merilis fitur validasi serupa.

Target Market



- **PPJK Swasta (Customs Brokers):** Otomasi audit kode HS dari 4 jam menjadi < 5 detik untuk mencegah salah deklarasi dan denda pabean.
- **Freight Forwarder & 3PL:** Lindungi kontainer klien dari penahanan Jalur Merah serta risiko denda sewa kontainer (Demurrage).
- **Importir & Korporasi Besar:** Validasi izin Lartas mandiri, jamin kerahasiaan data invoice secara on-premise, dan temukan potensi restitusi pajak.
- **Platform Logistik & Developer:** Integrasikan modul skrining pabean otomatis ke sistem ERP internal melalui API dan Model Context Protocol (MCP).

Kompetitor & Moat



The IDC MarketScape: Worldwide Global Trade Management for Retailers and Importers 2025-2026 Vendor Assessment

- **100% On-Premise:** Data rahasia harga beli pabrik dan invoice tersimpan aman di server lokal tanpa risiko bocor ke cloud pihak ketiga.
- **Spesialisasi Lartas Indonesia:** Memetakan langsung 5.612 kode HS serta 546 perizinan wajib kementerian teknis (SDPPI, Kemenkes, BPOM).
- **Kalkulator Demurrage CMA CGM:** Memodelkan estimasi denda penahanan kontainer harian secara presisi dan matematis berbasis tarif resmi pelayaran.
- **Deteksi Restitusi Lebih Bayar:** Proaktif mengidentifikasi kelebihan bayar bea masuk akibat salah tarif untuk diklaim kembali oleh importir.

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HACKTIV8

Thank You